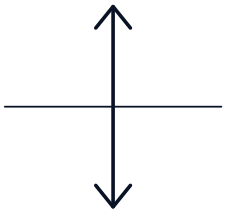
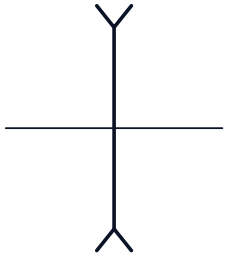
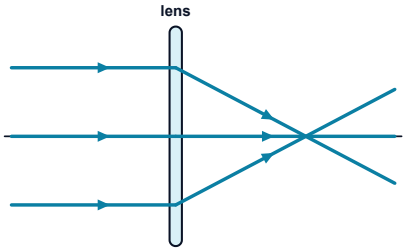
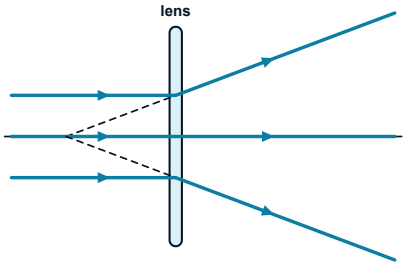
DO NOW, AO1 REVISION

Lenses and visible light

Waves
Paper 2, Separate Physics only
25 questions

Cover the answers. Answer each question out loud or on paper, then check. Tick the ones you know. Learn the 6 starred (★) questions first.

	QUESTION	ANSWER	✓
1	**Name the process that happens to light in a lens.**	Refraction	☐
2 ★	**Describe what a convex lens does to parallel rays of light.**	It brings them together at the principal focus.	☐
3	**What is the focal length of a lens?**	The distance from the lens to the principal focus.	☐
4	**Describe what a concave lens does to parallel rays of light.**	It spreads them out, as if from the principal focus.	☐
5 ★	**What is a real image?**	An image formed where rays of light actually meet.	☐
6 ★	**What is a virtual image?**	An image that rays of light only appear to come from.	☐
7	**Name the type of image that can be shown on a screen.**	A real image	☐
8	**Name the type of lens that produces only virtual images.**	A concave lens	☐
9	**Name the type of lens that this symbol represents.** 	A convex lens	☐

	QUESTION	ANSWER	✓
10	**Name the type of lens that this symbol represents.** 	A concave lens	☐
11 ★	**Name the type of lens shown in this ray diagram.** 	A convex lens	☐
12	**Name the type of lens shown in this ray diagram.** 	A concave lens	☐
13	**State how different colours of light differ.**	They have different wavelengths.	☐
14 ★	**What is specular reflection?**	Reflection from a smooth surface in a single direction.	☐
15	**What is diffuse reflection?**	Reflection from a rough surface, scattering light in many directions.	☐
16	**Explain how colour filters work.**	They transmit some wavelengths and absorb the rest.	☐
17	**Explain why blue objects appear blue.**	They reflect blue light and absorb the other colours.	☐
18	**Explain why white objects appear white.**	They reflect all colours of light equally.	☐

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	QUESTION	ANSWER	✓
19	**Explain why black objects appear black.**	They absorb all colours of light.	☐
20	**In words, state the equation for magnification.**	magnification = image height ÷ object height	☐
21	**State the unit of magnification.**	It has no unit (it is a ratio).	☐
22	**State the types of image that a convex lens can produce.**	Real or virtual	☐
23	**State what determines the colour of an opaque object.**	The wavelengths of light it reflects most strongly.	☐
24	**State what an opaque object does to light.**	It absorbs or reflects it; none is transmitted.	☐
25 ★	**A blue object is viewed in red light. State and explain the colour it appears.**	Black. It absorbs the red light and has no blue light to reflect.	☐

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